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AI-based novelty detection in crowdsourced idea spaces

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ABSTRACT

Processing large and heterogeneous numbers of ideas submitted to crowdsourcing contests is a regular challenge for idea evaluators. The aim of this study is to investigate a potential use case for AI-based innovation management and to extend the knowledge of using automated novelty detection in idea evaluation processes. AI-based language models can automatically allocate short texts according to their semantic similarity in an embedded space. We represent the semantic content of crowdsourced ideas with the three contemporary text embeddings – Doc2Vec, SBERT, and GPT-3-based Ada Similarity – and compute their semantic distance to different reference sets using different novelty detection algorithms. We then compare the algorithm-generated scores with human novelty assessments to validate them. While selected novelty scores based on text embeddings correlate with humans, our results show that scores based on SBERT embeddings best match human novelty assessments. We also find that AI-based novelty detection approaches perform better for ideas below the median word count and when compared to a set of existing solutions, suggesting that the chosen language model is not the only factor influencing the applicability of the proposed approach. Furthermore, the study highlights important features and limitations of automatically generated novelty scores that need to be considered when complementing evaluators searching for new ideas in crowdsourcing contests and beyond.

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Crowdsourcing; idea space; language model; text embedding; novelty detection; AI-based innovation management; idea evaluation

Introduction

Crowdsourcing platforms allow organisations to search for new solutions and accumulate massive amounts of ideas for their broadcasted problems (Jeppesen & Lakhani, 2010; Terwiesch & Xu, 2008). Compared to situations where organisations develop solutions within innovation departments, a larger solution space can be explored (von Hippel & von Krogh, 2016). However, for evaluators, it is challenging to deal with the enormous amount of generated input, grasp the potential of each suggestion and perspective, and select novel ideas to move their organisation ahead. Reading through all ideas requires substantial resources in terms of time and human resources and often leads to high

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cognitive effort and workload, negatively affecting the recognition of novelty (Criscuolo et al., 2017). The crowding of large pools of ideas (Piezunka & Dahlander, 2015) narrows the evaluators' scope of attention to a subset of suggestions, reducing the chance of identifying novel ideas distant from their previous knowledge. Furthermore, ideas unrelated to the market trends and established product categories (Abrahamson, 1991; Duan et al., 2009) or areas of expertise (Boudreau et al., 2016) are likely to be overlooked or underestimated.

To increase the understanding of potential solutions, scholars argue to move from a search over single idea representations to a search over a meta-representation (Newell & Simon, 1972). Insights into the meta-structure of a crowdsourced idea space may enable decision-makers to search for previously unknown possibilities. While former endeavours to reveal the structure of idea spaces relied on manual categorisations, keyword annotations (Westerski et al., 2013), or human similarity assessments (Kornish & Ulrich, 2011), today, AI-based language models enable an automatised allocation of idea texts according to their semantic meaning.

Previous innovation studies that computationally searched for novel content often applied bag-of-words models to represent the semantic meaning of ideas by their term frequencies or thematic topics (Toubia & Netzer, 2017; Wang et al., 2019) or text embeddings (Hao et al., 2017; Jeon et al., 2022). Text embeddings assign numeric values to text documents to compare their semantic similarity (Muennighoff et al., 2022; Naseem et al., 2021). By combining it with various distance-based novelty detection methods, such as the k-nearest-neighbour (Ramaswamy et al., 2000) and local outlier factor (Breunig et al., 2000), the semantic distances between the numeric representations of idea or technology descriptions served as a proxy to measure the novelty or uniqueness of ideas depending on the reference set. While these studies report promising results in identifying novel ideas or technologies, they also note that the results of their approaches depend on the algorithms chosen.

Accurate semantic representations are essential for novelty detection approaches comparing the semantic similarity of ideas. Therefore, innovation researchers and practitioners are well advised to consider advancements in available language models to better understand their applicability and potential for AI-based innovation practices (Bouschery et al., 2023; Füller et al., 2022). Recently, contextual language models based on deep learning transformer architectures and pre-trained on unprecedentedly large text data entered the stage. While transformer-based text embeddings for representing short documents, e.g., SBERT (Reimers & Gurevych, 2019) or GPT-3-based Ada Similarity (Neelakantan et al., 2022), set new standards in text similarity benchmark tasks, research has also shown that their performances vary depending on the tasks' contexts (Muennighoff et al., 2022).

Although transformer-based language models revolutionised almost any natural language processing task (Wolf et al., 2020), we know little about their potential for capturing idea novelty. According to a recent survey, a third of the responding AI-savvy organisations apply natural language understanding in products or business processes, whereas only eleven percent have experience with transformer-based models (Chui et al., 2022). Equally, innovation researchers have not yet considered transformer-based text embeddings for AI-based novelty detection, nor have they compared the capabilities of established text embeddings in representing ideas.

The interest in AI-based approaches for novelty detection is rooted in their expected ability to filter novel contributions from large sets of ideas by using algorithms that automatically analyse the collected content rather than relying on human processing skills. However, while the approaches promise to provide efficient and objective support for assessing novelty, innovation researchers' and practitioners' knowledge about the reliability of AI-based methods to capture the novelty of ideas is rather limited. This study aims to clarify the emerging field of AI-based novelty detection and explore its applicability for identifying novelty in crowdsourced ideas. More specifically, we compare different algorithm-generated novelty scores with human novelty evaluations in order to better understand the key determinants of our semantic similarity-based approach. Furthermore, it sheds light on the possibilities and current limitations of AI-based novelty detection in innovation management.

We represent the semantic content of 232 crowdsourced ideas with the contemporary text embeddings Doc2Vec, SBERT, and GPT-3-based Ada Similarity and apply established distance-based metrics like the k-nearest-neighbour and local outlier factor on data from a crowdsourcing contest particularly suitable for investigating novelty. In a validation step, we compare the algorithm-generated novelty scores to human novelty assessments of the same ideas. While selected measures of all investigated models for semantic text representation correlate with human novelty evaluations, we find that those relying on SBERT comply best with humans. Our analysis reveals that AI-based novelty detection works better for ideas below the median word length and when comparing crowdsourced ideas to a range of existing product categories. Furthermore, we analyse cases with strongly deviating novelty assessments of humans and algorithms to highlight potential peculiarities and limitations of AI-based approaches to novelty detection.

Our findings suggest that not only the choice of language model, but also the consideration of the frame of reference, e.g., crowdsourced or pre-existing ideas, and the features of the processed idea descriptions, e.g., length, style or thematic focus, are important to unlock the potential of AI-based novelty detection as a complementary toolkit in idea evaluation. While we discuss some limitations of the investigated AI-based novelty detection approach, such as shortcomings in capturing nuances in idea descriptions, processing unconventional structures and longer texts, or dependencies on pre-defined reference knowledge, we also outline possible interventions. The study informs and encourages the development of effective AI-based innovation practices to help decision makers identify and evaluate novel ideas collected in crowdsourcing and other ideation formats.

Literature background

Crowdsourced idea spaces and the recognition of novelty

Over the last years, crowdsourcing has gained momentum in organisations to solicit novel solutions to innovation problems from external and internal sources through online platforms (Jeppesen & Lakhani, 2010; Terwiesch & Xu, 2008). Organisations can accumulate valuable solution-related knowledge via crowdsourcing contests, including multiple ideas for new products or services. However, developing a clear picture of the extensive and often heterogeneous idea submissions is challenging for decision-makers. While they can draw

on a high domain-specific knowledge and experience level, reading through all ideas and sense-making requires significant time and cognitive resources. Moreover, when evaluators process a vast number of ideas with multiple interacting components, their processing capacities easily reach the limits, as too many ideas are delivered at a certain point in time, and the incoming information cannot be sufficiently organised and recognised by their thematic content (Cheng et al., 2020; Sweller, 2003).

Furthermore, the attention space of decision-makers in new product development is often built on existing market information, e.g., consumer needs, technology trends, or activities of competitors. Ideas outside the business mainstream that lie off the beaten path or are distant to own areas of expertise are likely to be overlooked (Boudreau et al., 2016). Evaluators compensate for their lack of information about the actual value of a new unknown product by inferring from previous actions (Duan et al., 2009). Such phenomena may explain why certain product categories or technologies dominate over more efficient but less popular ones (Abrahamson, 1991). In the worst case, decision-makers reject potential innovations without considering their attributes and benefits.

The tendency of evaluators to choose options close to the status quo increases with the number of options in the choice set (Samuelson & Zeckhauser, 1988). However, a sound evaluation requires an even-handed review of all available options. When all the generated ideas within a crowdsourcing contest are brought into the attention space of evaluators at once, the increased workload may limit evaluators' preferences to identify novel ideas distant from their existing knowledge stocks. As a result, organisations tend to proceed with projects with intermediate levels of novelty (Criscuolo et al., 2017; Piezunka & Dahlander, 2015).

In order to discuss possible ways to bring novel ideas into the attention space of evaluators, a clear understanding of the concept of novelty is required. Dean et al. (2006, p. 648) define novel ideas as rare and unusual and argue that an idea's novelty needs to be assessed 'in relation to how uncommon it is to the mind of the idea rater or how uncommon it is in the overall population of ideas'. Dahlin and Behrens (2005) analysed patent novelty and distinguished between novel and unique solutions. While novel inventions are dissimilar from prior ones, unique ideas must be distant from other currently available solutions. Furthermore, conceptualisations of novelty vary depending on the context in which it is assessed (Foster et al., 2022; Rosenkopf & McGrath, 2011). When translating the definitions to the context of crowdsourcing contests, an idea's degree of novelty is determined by comparing it to the knowledge of existing solutions in the minds of idea evaluators. Moreover, it can also be argued that the ideas shared in the crowd represent all currently available solutions, and one should seek outliers representing unique submissions in the crowdsourced idea space. In this context, the crowdsourced ideas under consideration and the knowledge about previously existing solutions represent their attention or solution space.

There are good arguments in innovation literature that novel ideas are underestimated in the idea evaluation process. They are likely to receive insufficient attention if they deviate from the business mainstream (Abrahamson, 1991), represent unknown categories (Boudreau et al., 2016; Duan et al., 2009), or enter the attention space of evaluators together with many other potential solutions (Piezunka & Dahlander, 2015). Therefore, evaluators must overcome cognitive challenges in processing extensive and heterogeneous idea content. AI may compensate for cognitive limits in humans' information processing capacities.

Leading innovation researchers are pushing for the future of innovation management to include AI allowing humans to free up resources or join forces to increase the efficiency and effectiveness of innovation processes (Cockburn et al., 2019; Füller et al., 2022). In particular, recent advances in natural language processing models attracted considerable attention in innovation research (Antons et al., 2020) and further led to a surge in the number and power of available language models.

AI-based language models for semantic text representation

As crowdsourced ideas are described in text form, processing them with AI-based language models allows for generating an automated similarity mapping of idea descriptions according to their semantic content. Structuring multiple ideas based on their feature similarity appears useful to better explore novel contributions, including those that might have been underestimated or undetected due to hurdles in current evaluation procedures.

In the last few years, the development of language models for semantic text representation advanced significantly (Naseem et al., 2021). Traditional bag-of-words models count word occurrences and convert them into numeric values. It is called a ‘bag’ of words because information about the order and structure of words in the document is not considered. Previously, several crowdsourcing researchers applied popular models such as tf-idf or topic modelling algorithms like LDA to analyse structures in the gathered knowledge (Kakatkhar et al., 2018; Toubia & Netzer, 2017; Wang et al., 2019). Thereby, documents are represented by sparse vectors encoding the statistical frequency of each word, e.g., tf-idf, or by latent topic distributions that are assigned to each document, e.g., LDA. However, when analysing a high number of documents, the models cannot effectively measure the proximity between the documents due to the sparsity in the vector representation (Kim et al., 2017). Many studies have shown that models based on neural networks outperform classic bag-of-words models in document similarity tasks (Chandrasekaran & Mago, 2021; Dai et al., 2015; Le & Mikolov, 2014).

Unlike bag-of-words models that are based on statistical distributions of single words, continuous models for text representation assign similar vector values to semantically similar words and capture synonymous meanings in language. Models such as Word2Vec (Mikolov et al., 2013) or GloVe (Pennington et al., 2014) rely on large datasets and neural network algorithms to learn semantic text embeddings for all unique words in a vocabulary depending on how frequently they appear close to each other in a text corpus. In the process, self-supervised learning approaches are applied to predict a word given its neighbouring context or vice versa. For example, in the innovation context, Lee et al. (2020) applied Word2Vec to construct a vectorised product space locating similar technologies close to each other. Not only similar words but also entire sentences or short documents can be located close to each other in an embedded space (Le & Mikolov, 2014). For example, the sentence ‘I gave an innovation talk at the university on the same date’ must have a similar semantic vector representation as ‘I had a new product development lecture at the research institute’. However, continuous models ignore the meaning of words in different contexts and cannot capture homonyms. For example, a processed word like ‘paper’ has different meanings depending on whether one writes academic articles or discusses packaging solutions.

Unlike continuous models, where each word in a vocabulary corresponds to a single word embedding, pre-trained contextual models consider the environment of each occurrence of a word. Thus, each embedded vector's representation differs depending on the sentence or paragraph. This advancement was facilitated by deep-learning transformer architectures, which allow learning universal representations of an unprecedented amount of textual data scraped from web corpora, e.g., Wikipedia, Reddit, Common Crawl, or Google News. For example, transformer-based BERT text embeddings are generated by predicting a masked word in a sentence given its neighbouring context and a process called self-supervised learning. The SBERT framework fine-tunes BERT embeddings to describe the semantic content of entire sentences or paragraphs with numeric vectors (Reimers & Gurevych, 2019). A recent comprehensive benchmarking study showed that SBERT models are highly competitive considering speed and accuracy in generating text embeddings (Muennighoff et al., 2022). These models outperform continuous ones in standardised benchmarking tasks measuring the overlap with human similarity assessments and semantic search performance. However, as SBERT models are fine-tuned based on similar sentences, how well they capture the content of entire idea descriptions remains unclear. In a recent blog post, OpenAI claimed that the newest version of their transformer-based Ada text embeddings that modifies a GPT-3 language model (Neelakantan et al., 2022) is 'convenient to work with long documents' (Greene et al., 2022).

AI-based novelty detection

Considering the functional relationship between novelty and similarity, we conceptualise the identification of novel ideas shared in crowdsourcing contests as a search over an idea space. While AI-based language models can reveal the similarity structure of possible solutions in a crowdsourced idea space, the automated identification of novelty based on semantic proximities between ideas requires complementary algorithms. In this case, novelty is measured by a configurational approach (Foster et al., 2022), which relies on the ability of text embeddings to represent ideas as semantic feature vectors.

Using density-based novelty detection algorithms (Carreño et al., 2020) seems obvious to identify novel submissions in crowdsourced idea spaces. Novelty detection in AI shares some central properties with definitions of novelty in innovation management. For example, new observations are compared to reference or training datasets they are part of, and novel ideas must be distant from existing ones serving as a reference dataset (Carreño et al., 2020). Similarly, innovation research argues that novelty is a function of proximity and should be evaluated according to how unusual or distant it is to the mind of the idea evaluator or the known population of ideas (Bavato, 2022; Dahlin & Behrens, 2005; Dean et al., 2006).

Several studies searching for new ideas in the front-end of innovation explored the potential of AI-based language models to represent the structure of idea spaces and look for novel or unique content. Toubia and Netzer (2017) applied tf-idf and topic modelling to represent ideas and measured the semantic distance between the idea vectors and an averaged baseline knowledge vector to explore the prototypicality of crowdsourced ideas. Wang et al. (2019) applied sparse tf-idf and topic model vectors for semantic text representation to compute the novelty of ideas generated by crowd workers and compare

it to novelty assessments of experts. Hao et al. (2017) implemented a continuous Doc2Vec model to represent design concepts and applied the k-nearest-neighbour cosine distance to measure their novelty. Recently, Jeon et al. (2022) combined a continuous Doc2Vec embedding with the local outlier factor to identify novel patents and found significant correlations with their measure and citation-based measures for the radicalness of patents.

Different models for semantic text representation can be combined with different distance-based novelty detection methods. Previous studies highlight the exciting potential of algorithm-supported approaches but also report mixed results. Despite identifying significant correlations between AI and human novelty ratings, they highlight the risk of missing truly innovative ideas by only following AI-based scores (Wang et al., 2019) and call for the adoption of more advanced language models (Jeon et al., 2022; Westerski & Kanagasabai, 2019).

Given the advancements in transformer-based text embeddings, new opportunities to better capture novelty in crowdsourced ideas have emerged in recent years. Jeon et al. (2022) already engaged in first experiments with contextual text embeddings and suggested that a continuous Doc2Vec model is superior for patent novelty detection. However, the calculated patent vectors were based on averaged BERT word vectors without considering the advancements in available embeddings for semantic text representation. Furthermore, text embeddings are usually evaluated on a small number of datasets for a single standardised task without considering their potential application context (Muennighoff et al., 2022). We know little about how different embeddings and conceptualisations for novelty detection compare and how well they can capture the novelty of ideas shared in a crowdsourcing contest. Moreover, studies in the field of crowdsourcing rarely distinguish between novelty and uniqueness when conceptualising AI-based approaches to support decision-making.

This study aims to clarify the emerging research field of AI-based novelty detection and explore its applicability to identify novelty in crowdsourced contributions. Treating novelty as a function of proximity (Bavato, 2022), we apply contemporary text embeddings and distance-based metrics on data from a crowdsourcing contest particularly suitable for investigating novelty. In this process, we want to discover whether the conceptualised techniques can capture novelty similarly to humans and learn more about their potential as a new AI-based innovation practice (Cockburn et al., 2019; Füller et al., 2022).

Although the automatic generation of novelty ratings based on the semantic content of crowdsourced contributions promises to lighten the load on humans processing large volumes of ideas and provide an objective basis for decision-making, the actual value of AI-based novelty detection is best determined in comparison to the results of human processes. Thus, we collect additional human novelty assessments of the crowdsourced ideas to validate the algorithm-generated novelty scores and learn more about their relationships to human novelty assessments. Given the findings on biases against novel ideas (Boudreau et al., 2016; Criscuolo et al., 2017; Piezunka & Dahlander, 2015), here, one may also ask whether it is even necessary or desirable that algorithm-generated scores match human novelty evaluations and diverging assessments may complement decision-makers in the process. In the research process, we not only explore whether and how well the different algorithm-based approaches can capture novelty but also try to reveal the peculiarities and limitations of such approaches in complementing the identification of novel ideas in crowdsourcing contests.

Methodology

Data source

A global hygiene and health organisation selling household towels reached out to an external crowd at a leading German innovation platform in November 2020. In the crowdsourcing contest, the organisation wanted to know more about customers' pain points in using household towels and asked for solutions around the towel of the future. Within four weeks, the crowd generated 232 potential solutions. The participants could see and comment on each other's ideas, and they had the chance to win prizes of € 3,000 in total.¹

We have chosen this contest for its particular suitability to explore the applicability of algorithm-augmented approaches for novelty detection. The global hygiene and health company explicitly stated ten previously existing product groups in the contest briefing. This allows us to empirically measure each crowdsourced solution's novelty relative to the company's existing products. As an alternative data source for a reference set of previous solutions (Toubia & Netzer, 2017), we scraped all paragraphs of weblinks appearing on a Google search's first five result pages with the input 'How could paper towels be improved?', yielding 1,131 paragraphs. Furthermore, right after the contest, four employees of the hosting company including the global brand director and three innovation managers self-selected 58 most interesting submissions of the contest participants and evaluated their novelty using a five-point Likert scale (1=not novel, 5=novel).² They based their evaluation on the straightforward question 'how novel is the use case for [company name]?' using their knowledge on existing products at that time. Though the sample may be too small to derive significant relationships in most cases, the data helps to check the validity of the novelty assessment of other sources.

Algorithm-based novelty score generation

We implemented three contemporary text embedding models on the data from the crowdsourcing contest to generate a vectorised representation of the crowdsourced idea space. Using the Python package *Gensim* (Rehurek & Sojka, 2010) we generated Doc2Vec embeddings to represent the semantic content of each idea in a numeric vector. The model was trained on text from all idea descriptions, the set of existing products, and the scraped paragraphs from the Google search results, adding up to 1289 documents. We experimented with vector sizes of 64, 128, and 256 with window sizes of 5, a minimum word count of 2, and 50 training epochs, following Jeon et al. (2022). In the result section, we report the Doc2Vec model outputs with 256 dimensions, showing the highest correlation with the human novelty evaluation.

The Python package *Sentence-Transformers* involves an extensive collection of pre-trained SBERT models (Reimers, 2021). We applied the pre-trained SBERT embedding 'all-mpnet-base-v2' with 768 dimensions – the leading model in the semantic similarity benchmark – to calculate idea vector embeddings for each existing solution, scraped paragraph, and each idea submitted to the crowdsourcing contest. We experimented to find the highest possible threshold of processable word tokens and set the maximum length to 512.³

The Python package *OpenAI* (OpenAI, 2023) and a corresponding API allowed us to generate Ada Similarity embeddings. We used their latest version ‘text-embedding-ada-002’ to model idea vectors for each existing solution, scraped paragraph, and each idea submitted to the crowdsourcing contest. The embeddings have 1,536 dimensions and can process a context length of 8,191 input tokens.

To measure the distances between existing household towels of the company and the representations of the crowdsourced solutions in the four vectorised idea spaces, we applied the k-nearest-neighbour and local outlier factor measures using the Python Package *scikit-learn* (Mueller, 2021). The k-nearest-neighbour measure determines novel observations by their distance to the k^{th} nearest point in the reference dataset at a numeric scale (Ramaswamy et al., 2000). Consequently, the novelty score of each crowdsourced idea vector is calculated by the cosine distance to the k^{th} closest vector representing existing products of the company or the scraped paragraphs from the Google search. The local outlier factor is another nearest-neighbour-based measure to identify novel observations using a numeric scale (Breunig et al., 2000). The term local refers to its ability to identify isolated entities relative to its surrounding neighbourhood. Thus, it can filter out outliers in densely-populated local neighbourhoods that other approaches cannot identify.

For all generated idea embeddings, we calculated nine different algorithm-based novelty scores, the first- and third-nearest-neighbour distance and the local outlier factor, relative to the set of existing products, the set of scraped Google search paragraphs, and first- and the other crowdsourced ideas.⁴

Human validation sets

We created two additional validation sets to determine whether and how well the generated scores can capture novelty, apart from the evaluations of the company’s experts hosting the crowdsourcing contest. The first one involved an evaluation of the crowdsourced ideas by three innovation researchers with expertise in crowdsourcing and idea evaluation.⁵ They were unaware of the algorithm-generated novelty scores. Before their evaluation, they read through all existing solutions stated by the global hygiene and health company and screened the web pages of their towel brands. The novelty of the ideas was rated on a seven-point Likert scale (1=strongly disagree, 7=strongly agree). Before and during the evaluation process, they exchanged their understanding of novelty, agreeing on Dean et al.’s (2006) definition of rare and unusual solutions. The intraclass correlation coefficient was 0.51, a satisfactory value for the chosen seven-point scoring scheme and the individual room for interpretation in evaluating novelty. We proceeded with the mean evaluation for each idea to compare the algorithm-generated scores.

The second validation set stems from crowd workers acquired from the online research platform Prolific. Research has shown that participants from this platform tend to provide the most honest answers compared to other platforms (Eyal et al., 2021). As we could not screen for participants working in fields related to household towels, we tried to constrain our sample on the customer side. Users can be appropriate proxies for experts when evaluating new product or service ideas (Magnusson et al., 2016). We sampled English-speaking female participants above the age of 25, who are

most likely to use household towels frequently.⁶ In a short online survey, each participant had to read through all existing solutions stated by the global hygiene and health company and answer a comprehension question integrated into the subsequent evaluation process. Then they evaluated the novelty of three randomly assigned crowdsourced ideas about future household towels⁷ on a five-point Likert scale (1=strongly disagree, 5=strongly agree). 773 participants completed the tasks, from which 577 fulfilled all implemented attention and comprehension checks. On average, each idea in the final validation set was evaluated 8.53 times. We used the mean evaluation for each idea to compare the algorithm-generated scores.⁸ The descriptive statistics about all three human novelty scores can be found in Table A1 in the Appendix, while Table A2 provides examples of evaluated ideas.

Results and findings

Correlation analyses

To validate the different approaches and to examine patterns between the different novelty scores, we performed pairwise correlation analyses using Spearman’s rank order method. First, we compared the assessments of the expert employees right after the crowdsourcing contest with the other validation sets generated by the three researchers and the crowd workers from Prolific (Table 1). Second, we calculated the pairwise correlations between algorithm-generated novelty scores and human assessments (Table 2). In addition, as we were interested in potential differences in the models’ capabilities to generate high-quality semantic representations at different text lengths, we also analysed the correlations within subsets of shorter and longer ideas (Table 3). Therefore, the initial set of 232 ideas was split at the median length of 128 words. In the following, we discuss the results in more detail.

Human validation sets

The novelty evaluations from all three sources – company experts, researchers, and crowd workers – are all significantly positively correlated with each other and stable across different idea lengths. On the one hand, this confirms the suitability of the different validation sets. On the other hand, it is a first indication that an utterly consistent interpretation and capture of novelty is challenging. However, the strong and significant correlation between the employees of the global hygiene and health company and the novelty evaluations of the researchers is higher than with the crowd workers. This gives confidence about the validity of the novelty evaluations of the researchers and shows that crowd workers perceived novelty differently.

Table 1. Pairwise correlation human novelty ratings.

	Expert	Researcher	Prolific
Expert (n = 58)	1		
Researcher (n = 225)	0.60***	1	
Prolific (n = 203)	0.34*	0.55***	1

* $p > 0.05$, ** $p < 0.01$, *** $p > 0.001$.

Table 2. Pairwise correlation of algorithm-generated novelty scores with human ratings.

	1st NN company	1st NN scraped	1st NN unique	3rd NN company	3rd NN scraped	3rd NN unique	LOF company	LOF scraped	LOF unique
Doc2Vec									
Expert (<i>n</i> = 58)	0.09	0.09	0.02	−0.09	0.01	−0.12	−0.07	−0.03	0.09
Researcher (<i>n</i> = 225)	0.23***	0.14*	−0.07	0.19**	0.10	−0.09	0.20**	0.11	−0.03
Prolific (<i>n</i> = 203)	0.08	−0.07	−0.03	−0.01	−0.10	−0.10	0.06	−0.04	−0.14
SBERT									
Expert (<i>n</i> = 58)	0.15	0.19	−0.08	0.24	0.16	0.12	0.19	−0.12	0.16
Researcher (<i>n</i> = 225)	0.25***	0.32*	0.09	0.32***	0.32***	0.27***	0.29***	0.13	0.25***
Prolific (<i>n</i> = 203)	0.07	0.13	0.05	0.08	0.12	0.07	0.08	0.21**	0.07
GPT-3 Ada Similarity									
Expert (<i>n</i> = 58)	0	0.13	−0.05	0.12	0.13	0.12	0.04	0.02	0.07
Researcher (<i>n</i> = 225)	0.15*	0.21**	0.01	0.25***	0.20*	0.22**	0.23**	0.18**	0.14*
Prolific (<i>n</i> = 203)	0.03	0.14*	0.08	0.07	0.08	0.04	0.07	0.15*	0.05

* $p > 0.05$, ** $p < 0.01$, *** $p > 0.001$; NN = nearest neighbour; correlations between all algorithm-generated scores in Table A3.

Table 3. Pairwise correlation of algorithm-generated novelty scores with human ratings (shorter vs. longer ideas).

	1st NN company	1st NN scraped	1st NN unique	3rd NN company	3rd NN scraped	3rd NN unique	LOF company	LOF scraped	LOF unique
Doc2Vec shorter									
Expert (<i>n</i> = 23)	0.22	0.32	0.03	0.06	0.24	−0.25	0.09	0.11	0
Researcher (<i>n</i> = 112)	0.26**	0.2*	−0.05	0.23*	0.09	−0.12	0.26**	0.15	−0.06
Prolific (<i>n</i> = 100)	0.17	−0.03	0.07	0.06	−0.08	−0.19	0.17	0.02	−0.19
Doc2Vec longer									
Expert (<i>n</i> = 35)	0.04	0.06	0.01	−0.12	0.06	−0.07	−0.14	−0.12	0.09
Researcher (<i>n</i> = 113)	0.21	0.06	−0.09	0.16	0.11	−0.05	0.15	0.09	0.02
Prolific (<i>n</i> = 103)	−0.03	−0.12	−0.13	−0.09	−0.13	0.01	−0.05	−0.12	−0.07
SBERT shorter									
Expert (<i>n</i> = 23)	0.2	0.33	0.21	0.44*	0.27	0.33	0.34	0.16	0.23
Researcher (<i>n</i> = 112)	0.39***	0.43***	0.1	0.44***	0.44***	0.34***	0.45***	0.25**	0.29***
Prolific (<i>n</i> = 100)	0	0.08	0.02	0.09	0.08	−0.04	0.11	0.27**	−0.04
SBERT longer									
Expert (<i>n</i> = 35)	−0.07	0.01	−0.17	−0.06	−0.04	−0.15	−0.03	−0.29	0.02
Researcher (<i>n</i> = 113)	0.11	0.22*	0.08	0.21*	0.20*	0.23*	0.17	0	0.25**
Prolific (<i>n</i> = 103)	0.16	0.17	0.07	0.08	0.15	0.19	0.05	0.16	0.18
Ada Similarity shorter									
Expert (<i>n</i> = 23)	0.21	0.38	0.16	0.39	0.31	0.34	0.19	0.14	0.25
Researcher (<i>n</i> = 112)	0.19*	0.22*	0.1	0.30**	0.21*	0.23*	0.32**	0.18	0.16
Prolific (<i>n</i> = 100)	−0.04	0.09	0.05	0.02	−0.02	−0.06	0.04	0.07	−0.07
Ada Similarity longer									
Expert (<i>n</i> = 35)	−0.11	−0.07	−0.08	−0.11	−0.03	−0.14	−0.09	−0.08	−0.13
Researcher (<i>n</i> = 113)	0.1	0.20*	−0.11	0.19*	0.18	0.2*	0.13	0.17	0.16
Prolific (<i>n</i> = 103)	0.11	0.20*	0.1	0.12	0.17	0.1	0.1	0.23*	0.14

* $p > 0.05$, ** $p < 0.01$, *** $p > 0.001$; NN = nearest neighbour.

Although consistently reporting positive correlations to a substantial extent, the results indicate no significant correlations between the algorithm-generated novelty scores and the 58 ideas rated by the experts. Only one significant correlation between algorithm-generated scores and the expert ratings of short idea descriptions could be found. This is not surprising, as the sample of employee expert ratings may be too small to allow statistical significance. Algorithm-generated scores best comply with researchers' novelty ratings. Only three scores are significantly positively correlated with consumer ratings of crowd workers from Prolific.

AI approaches

Without distinguishing between the length of ideas, transformer-based embeddings, especially SBERT, best match human novelty ratings. The SBERT and Ada Similarity approaches show many significant positive correlations. While the results indicate less significant correlations for the SBERT approaches than for the Ada approaches, their magnitude consistently exceeds that of the other approaches. The results based on Doc2Vec occasionally show significant positive correlations. For all embeddings, the third-nearest-neighbour scores show the highest convergence.

Reference sets

Algorithm-based novelty scores that measure the distance of the crowdsourced ideas from the host company's existing product descriptions and the scraped paragraphs from a Google search asking for paper towel improvements best match human novelty ratings. Almost all scores show a significant positive correlation with the researchers' ratings. The uniqueness scores, which measure semantic distances to other submitted ideas in the crowdsourcing contest, show lower agreement with human novelty ratings compared to the other reference sets. AI-based uniqueness scores correlate positively and significantly with human ratings only when a sufficiently large neighbourhood is considered.

Idea length

When performing correlation analyses for the subset of ideas below and above of the median word count, our results show that most AI-based scores for ideas with fewer words have higher correlation coefficients. Several coefficients increase by ten to twenty percentage points compared to the analysis results of all idea lengths and even more compared to the subset of ideas longer than the median word count. Again, Doc2Vec shows the lowest agreement in several comparisons. Agreement between SBERT scores and human judgements increases significantly for shorter ideas, yielding higher coefficients than others. Despite the small sample size of expert ratings made immediately after the contests, the positive relationship between the novelty score, which measures the semantic distance between the SBERT embeddings of shorter ideas and those of existing solutions for kitchen towels, is statistically significant, with a coefficient of 0.44. The correlations between Ada novelty scores and human ratings are more stable than those of the other models when comparing the subset of short and longer ideas, at least for the ratings of researchers and crowd workers. Nevertheless, the coefficients of the subset above the median length remain weak and similar to those of SBERT embeddings.

Summary

The results show that the AI-based novelty detection outputs from the SBERT embeddings are most strongly related to human novelty assessments, especially when processing shorter idea descriptions and comparing them to the researcher rating. Several AI-generated novelty scores that rely on the other text embeddings Doc2Vec and Ada Similarity also converge with human assessments but with less consistency and magnitude. Although all human ratings are significantly positively correlated to each other, our results suggest that they assess novelty differently, leading to different agreement levels with AI-based scores. Keeping in mind that the true novelty of an idea is a concept challenging to fully ascertain, and the correlation between expert employees and

researchers reached coefficients of 0.6, our results provide evidence for the applicability of specific algorithm-generated scores to identify novel contributions in the examined crowdsourcing contest. Nevertheless, the coefficients remain at mediocre levels, around 0.3 for all and 0.4 for shorter ideas submitted to the contest. Therefore, additional analysis steps of high-discrepancy cases are worthwhile to better understand potential limitations and learn more about the relationship between human- and algorithm-generated novelty scores.

Analysis of high discrepancy cases

In an additional analysis, we qualitatively examined ideas with strongly deviating novelty assessments between humans and algorithms. Therefore, we measured the absolute difference between the third nearest neighbour SBERT and the researchers' novelty scores, indicating the highest significant positive correlation when considering ideas of all lengths. Then, we identified the top-ten percent of ideas with the highest discrepancy. This analysis step aims to identify patterns and better understand the peculiarities of AI-based novelty detection. For the comparison, all scores were normalised. In the following, we present the main results of the analysis of 22 extreme cases, dividing them into instances where the algorithm-generated score is higher than the human score and cases where the algorithm-generated score is lower than the human score. Of the 22 ideas, 63.6% belong to the former group and 36.4% to the latter. We did not detect differences in text length in the two subsets.

Algorithm-generated scores higher than human novelty scores

Sometimes, the algorithm-based scores assign much more novelty to ideas that describe related services like social donations of household towels or marketing programmes. The higher scores were not surprising as such solutions were not the focus of the crowdsourcing contest and were not part of the set of existing products stated by the company. Overall, often solutions that already exist but were not part of the reference prior knowledge or were transferred to a different application scenario received much higher novelty scores from the machine than humans.

In two cases, the machine may have struggled with the structure of the text, including short descriptions only describing pain points and solutions in a few key points using bullet points or other special signs. Another high-discrepancy idea includes a high amount of descriptive content with insufficient focus on the actual solution. As a result, humans evaluated all these ideas as common while receiving higher novelty scores from the algorithm. This pattern points out an obvious limitation: algorithms are likely to overweight unfamiliar content that uses unconventional text structure or describes the problem or context of an idea at the expense of evaluating the solutions' content per se.

The most characteristic pattern in the analysed sub-samples is the tendency of algorithm-based novelty scores to assign much higher novelty to ideas describing the application of an existing product in a different context which innovation researchers once called component control template (Goldenberg & Mazursky, 2002). The natural challenge of interpreting and ascertaining novelty became apparent when searching for patterns among the ideas with strongly deviating novelty scores. For example, wet tissues

are common products. However, when adding another healthy substance or fragrance and suggesting a new application or customer focus, one may assess it as more novel, but one could also argue that such a product already exists. For example, in one idea, a participant suggested the increased application of disinfection wipes in cars in response to the increasing demand for car-sharing services. Another idea suggests using one-time-use towels with refreshing effects for construction work and the provision of special containers.

Algorithm-generated scores lower than human novelty scores

Although the set of high-discrepancy ideas covers a wide thematic range, one specific theme around smart towels deserves a closer look. In the analysed subsample, two ideas deal with applying smart towels to signal pathogens by changing colour, using smart voice commands to control dispensers, or collecting health data synchronising with a smartphone. While the human raters evaluated all the smart towel solutions as highly novel, the algorithm shows much lower novelty scores. Reading through the two ideas' first-nearest-neighbours revealed more insights into why the actual novel ideas may not have been evaluated as that novel by the machine. The ideas matched with existing solutions that address the tasks they aim to innovate, e.g., solutions suggesting using different towel colours to distinguish their application field easier, using nicely designed dispenser boxes, or general ways to avoid potentially infected surfaces. Here, the SBERT language model and the semantic distance measures could not recognise the smartness in the proposed solutions and captured too much information on the problems they solve or other ideas that use different colours for other purposes. These cases suggest a possible limitation of the automatised approach in capturing small but important nuances within the idea content.

We all know paper towels are made of cellulose of plant origin. Looking at another example, one high-discrepancy idea suggests a replacement with cellulose of bacterial origin and argues that it increases water-holding ability. A small change in a major idea aspect can be interpreted as more or less novel. In our dataset, humans assigned much more novelty to substitute plant-based cellulose with bacterial one. We observed two more ideas characterised by replacement patterns (Goldenberg & Mazursky, 2002) among the eight ideas where machines assign less novelty to the corresponding.

Discussion

This work explores a potential AI-based innovation management practice (Bouschery et al., 2023; Cockburn et al., 2019; Füller et al., 2022) and extends the knowledge of how to integrate algorithm-generated novelty scores in idea evaluation. When evaluating large sets of ideas in crowdsourcing contests, AI-based novelty detection presents fast and straightforward access to particularly distinctive contributions. It automatically structures the idea space and identifies novel entities relative to any relevant reference set based on semantic text content in a few seconds without substantial costs or cognitive effort. The approach can support human evaluators to overcome cognitive limitations in information processing (Crisciuolo et al., 2017; Piezunka & Dahlander, 2015) and non-novelty biases (Abrahamson, 1991; Boudreau et al., 2016; Duan et al., 2009).

Our study contributes to the emerging research stream on AI-assisted ways to support decision-makers by computationally analysing novelty in the front-end of innovation (Hao et al., 2017; Jeon et al., 2022; Wang et al., 2019). While others suggested continuous Doc2Vec to identify novel solutions in patent spaces (Jeon et al., 2022), our results indicate that pre-trained contextual SBERT models are more suitable for measuring novelty in crowdsourced idea spaces (see Appendix for more details on the model). Unlike corpus-based models such as Doc2Vec, which are trained on the retrieved text documents, contextual language models are pre-trained on much larger datasets and can generate universal idea vectors that capture more context. Thus, they can provide a more meaningful representation of ideas in rather smaller datasets, such as the one examined in this study or various other ideation formats.

We also found that the choice of language model is not the only critical factor in the approximation of human novelty judgements by AI approaches. This adds important insights for the conceptualisation of configurational approaches to computationally measuring novelty (Foster et al., 2022). For example, decisions about the reference frame or the length of processed text can significantly affect algorithm-generated novelty scores, making consideration of context and input a key to unlocking the value of language models and related algorithms as a complementary toolkit in idea evaluation.

The concept of novelty describes how unusual or distant an idea is to the mind of an evaluator or a known reference set of solutions (Bavato, 2022; Dahlin & Behrens, 2005; Dean et al., 2006). In contrast to studies that approximate novelty with uniqueness in a set of crowdsourced ideas or patents, we propose to measure novelty relative to a pre-defined set of existing solutions. The finding that these scores comply best with human novelty evaluations independent of the applied text embedding confirms the usefulness of the conceptualisation.

AI-based approaches cannot initially exclude proposed solutions that humans with sufficient prior knowledge consider common. Because the company hosting the crowdsourcing contest referenced existing product categories in its briefing on the platform, we could use the data to define a reference set for the AI-based novelty detection approach. However, this reference set or scraped paragraphs from online searches only approximates what is known about existing household towel products. More comprehensive and meaningful datasets describing knowledge about previously existing solutions will improve the value of the proposed approach. Thus, before evaluators in organisations apply our customisable approach, a careful definition of the current knowledge on existing solutions to the broadcasted problem is required. Furthermore, by choosing the reference dataset, one can adjust the scope of the novelty they aim for, e.g., new to the company, community, or industry. This step directly influences the novelty scores and shows the versatility of the proposed approach.

The general pattern that algorithm-generated novelty scores of shorter ideas match human novelty scores better than those of longer ideas holds for all text embeddings. We conclude that the limiting reason why AI-based novelty detection is inferior for more extended ideas may not be solely explained by the quality and training process of the semantic text representation but also by the content and style of the idea descriptions. We found evidence of the risk of generating incorrect AI output by overweighting content that does not contain essential information about an idea or describes it in unconventional forms. The negative impact of the lack of algorithmic distinction between the

substance of an idea and its morphological representation can take different directions. Novelty may not be detected if novel aspects of an idea are only mentioned in single words or phrases and the rest of the description covers general knowledge similar to existing solutions. On the other hand, crowdsourced submissions with an unconventional structure or a significant amount of non-idea but distinctive text could result in a higher novelty score generated by the algorithm, even though the solution itself is already known. This limitation is likely to play a greater role in analysing the novelty of user-generated content in crowdsourcing competitions or online communities than, for example, patents that are described in a more standardised way.

While this implies that AI-based novelty detection may be susceptible to being deceived by stylistic devices and suffers from treating each processed word token equally by default, there are means to potentially overcome these limitations, including the use of AI. One option is to standardise the length of the idea descriptions submitted to crowdsourcing contests or directly ask for idea summaries in the submission process containing the essential aspects. Another option is to split longer idea descriptions into short paragraphs, encode them as individual text embeddings, and use the novelty score of the paragraphs with the closest semantic distance to the reference set as a measure of the whole idea. Where such interventions are impossible or have other drawbacks, AI-based language models can offer their services. For example, models for automatic text summarisation create shorter versions of a document while preserving critical information. While extractive methods cut out important segments from the original text and combine them into a summary, abstractive models build summaries from scratch without being forced to reuse phrases from the original text (El-Kassas et al., 2021). However, summarisation cannot guarantee that the core aspects of an idea remain in the condensed version. Thus, another option is fine-tuning language models based on text chunks labelled as idea or non-idea content to predict relevant content (Wolf et al., 2020; Zhang et al., 2021). More research on the possible positive effects of the mentioned interventions in the context of AI-based novelty detection is required.

The analysis of ideas characterised by strongly deviating novelty assessments between humans and algorithms revealed some important limitations in evaluating novelty with AI. We observed multiple reasons and patterns that may be responsible for considerable differences in the novelty assessments. For example, deficiencies in capturing nuances in the content became apparent. Moreover, substantial discrepancies can fall back on text descriptions that address the actual novel solution in only a small fraction of the description or on nuanced differences in text, e.g., a specific word or concept, with a strong impact on the novelty perception of ideas. These patterns suggest that novelty originating from replacing single product features (Goldenberg & Mazursky, 2002) is less likely to be captured automatically. While we investigated only a few extreme cases, future studies may extend the qualitative analysis of deviant cases and relate them to common idea types and templates.

In some cases, it is not so clear where the reasons for the divergent assessments lie. This finding confirms the frequently mentioned facet that a universal or true appreciation of the novelty of an idea is difficult to fully ascertain (Bavato, 2022; Foster et al., 2022). For example, is an idea novel simply because someone applies a known solution in a slightly different context? Or does the recombination create an entirely new solution with a new value, or is it just a concatenation of two existing solutions? Whose assessment of novelty is

correct? This challenge may also be reflected in the significant but different positive correlations between the three human validation sets.

Our findings suggest that full automation of the evaluation task is not recommendable. Instead, in line with similar research (Jeon et al., 2022), bundling capabilities and augmenting human novelty assessments with algorithm-based novelty scores seems useful. Although AI-based novelty detection should not be used to select ideas without human interference, it can extend the attention space of evaluators and support them in appreciating and selecting novel ideas through shortlisting particularly novel submissions. Focusing on identified subsamples with higher novelty or uniqueness values is advisable rather than filtering out single ideas based on their novelty. For example, novel or unique ideas in crowdsourcing contests can be shortlisted using meaningful thresholds and provide nuanced perspectives on the contributions. Established outlier thresholds can bring highly distinctive content into their attention space and counteract possible biases towards well-known solutions (see [Figure A4](#) in the Appendix). This can be of great use not only when analysing large volumes of ideas generated in crowdsourcing contests but also in related contexts such as market analysis, innovation workshops, or brainstorming sessions. While the proposed approach allows the computational measurement of the uncommonness and remoteness of an idea based on its semantic features, it cannot assess its cleverness – the third pillar of originality, which is a major component in assessing the creativity of an idea (Wilson et al., 1953). Future research on AI-based idea evaluation is encouraged to explore ways to incorporate the missing cleverness aspect, as well as the second creativity dimension of feasibility (Runco et al., 2012).

Furthermore, to some extent, discrepancies between human and algorithm-based novelty assessments could also be seen as an opportunity, especially in light of human biases against novelty. For example, AI-based approaches are more likely to identify ideas as novel when a known product is applied in another external context. Consequently, the deviating outputs could benefit evaluators looking for novel use cases of existing products or components. Nevertheless, we require more empirical research with more datasets to better understand the relationship between human- and AI-based novelty evaluations and to confirm the potential benefits of deviance. While our study focused on conceptualising and validating AI-based novelty detection, future research should also investigate the integration of such AI tools in organisational practice to determine the requirements for facilitating idea evaluation.

Notes

1. We checked for a relationship between the order of creation and the novelty and could not detect any significant correlation with our novelty measures. We also declare that we have taken ethical and legal aspects into account when collecting and processing data.
2. Most of the self-selected ideas were evaluated by one employee, while nine ideas were evaluated twice and two ideas were evaluated three times. We were not involved in this evaluation process and received the dataset at a later stage. This inconsistent evaluation process was also a major reason why we included the additional researcher validation set.
3. Please note that tokens are shorter than words. In our sample, five crowdsourced ideas exceeded the threshold of 512 word tokens and were truncated at this length.

4. As a distance metric we applied cosine similarity and set the number of neighbours to ten for the measure relative to the scraped paragraphs. As the existing product categories and the crowdsourced ideas are smaller datasets, we used five neighbours to calculate the local outlier factor.
5. One of them is also an author of this manuscript. Seven crowdsourced ideas were not rated by all three researchers due to lack of coherence in the content and are therefore excluded.
6. Although distribution slowly become more equitable, according to a recent survey data in the United States – a country from which respondents were sampled in addition to the United Kingdom – women still do far more housework than men (Brenan, 2020). Among heterosexual couples living together, 51% said the woman is more likely to clean the house, while only 9% said the man is more likely, with others reporting an even split or have no opinion.
7. Prior to the data collection we screened all 232 crowdsourced submissions for comprehensibility and excluded 29 submission for reason such as unclear descriptions and very long text including irrelevant information. Crowd workers may have a limited attention span compared to expert evaluators and comprehension issues are likely to negatively affect the quality of the novelty assessments.
8. The standard deviations of the ratings for all ideas are within a 95% confidence interval of (0.505, 1.578). The median of all standard deviations is 0.98. This means that on average 7 out of 10 ratings per idea differ by one Likert scale step.

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No potential conflict of interest was reported by the author(s).

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Appendix

SBERT in more detail

Transformer network architectures rely on an attention mechanism to relate different sequence positions to compute a representation of the text input and output, making the usage of recurrent and convolution neural networks unnecessary (Vaswani et al., 2017). While previous language models using deep learning learned their representation in only one direction, as a multi-layer bidirectional transformer encoder, BERT (Devlin et al., 2019) considers both the left and right context of text for generating the language model. Based on the original implementation of Vaswani et al. (2017), its input embeddings summarise the token embeddings, the segmentation embeddings, and the position embeddings (Figure A1). This allows for capturing the context of the text better. To train the bidirectional representation, some percentage of input tokens are randomly masked and predicted.

As training semantic representations of sentences and paragraphs is difficult and computationally inefficient with a classic BERT network, Reimers and Gurevych (2019) modified the model with a pooling operation to obtain a fixed size sentence embedding. In the modification or fine-tuning process it applies pairs of similar sentences and siamese and triplet network structures (Schroff et al., 2015) to update the weights of each embedded sentence via an objective function and derive semantically meaningful embeddings for comparison (Figure A2). The resulting SBERT models are suitable for large-scale semantic similarity comparison, clustering, and information

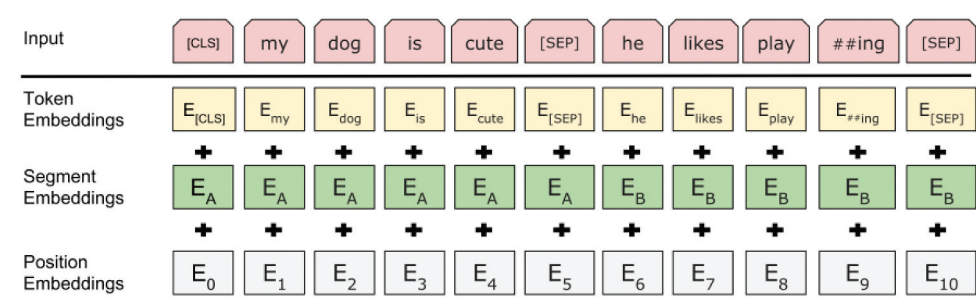


Figure A1. BERT input representation (Devlin et al., 2019).

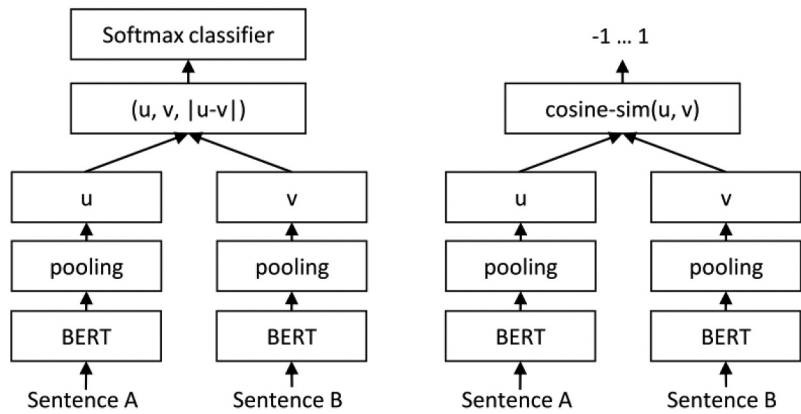


Figure A2. SBERT architecture with classification objective function (left) and similarity regression objective function (right), u and v represent the sentence embeddings (Reimers & Gurevych, 2019).

retrieval via semantic search. A comparison across benchmark tasks shows that they outperform former approaches that average BERT or GloVe embeddings.

SBERT models been trained for different purposes and evaluated for their quality in standardised benchmarking tasks for sentence embedding and semantic search performance. Depending on the size and pre-training approach SBERT model has 384, 512, 768, or even 1024 dimensions meaning that a vector with those dimensions represents a text (Reimers, 2021). Notably, the SBERT models truncate input text at a certain threshold. This threshold may be set differently. If ideas exceed reasonable thresholds, one could consider using other language models to summarise the idea text.

For an extensive instruction manual on how to use SBERT to compute semantic distances of text in research and practical settings, we advise studying the documentation and code examples in the referenced *Sentence-Transformers* Python package here:

https://www.sbert.net/docs/usage/semantic_textual_similarity.html#.

Documentation and guideline around OpenAI's Ada Similarity embedding can be found here: <https://beta.openai.com/docs/guides/embeddings>.

If readers are interested in our Python code and want to try our approach with their own datasets, we are happy to share our Google Colab notebook via Github here: <https://github.com/ThomasStroehle/Novelty/>.

Innovation researchers and practitioners that want to keep up with the latest developments and compare different text embeddings may also want to check the MTEB (Massive Text Embedding Benchmark) (Muennighoff et al., 2022) resources via <https://huggingface.co/blog/mteb>.

Representation of crowdsourced idea space

In Figure A3, we map four different hierarchical clusters and plot the corresponding idea titles of a cut-out of the generated SBERT representation of crowdsourced idea space. Therefore, we transformed the idea vectors from 768 dimensions into a two-dimensional space using UMAP dimensionality reduction (McInnes et al., 2018). A quick qualitative analysis of selected ideas shows how semantically similar ideas are accurately mapped in a simplified illustration of the idea space. For example, the dark green cluster on the top

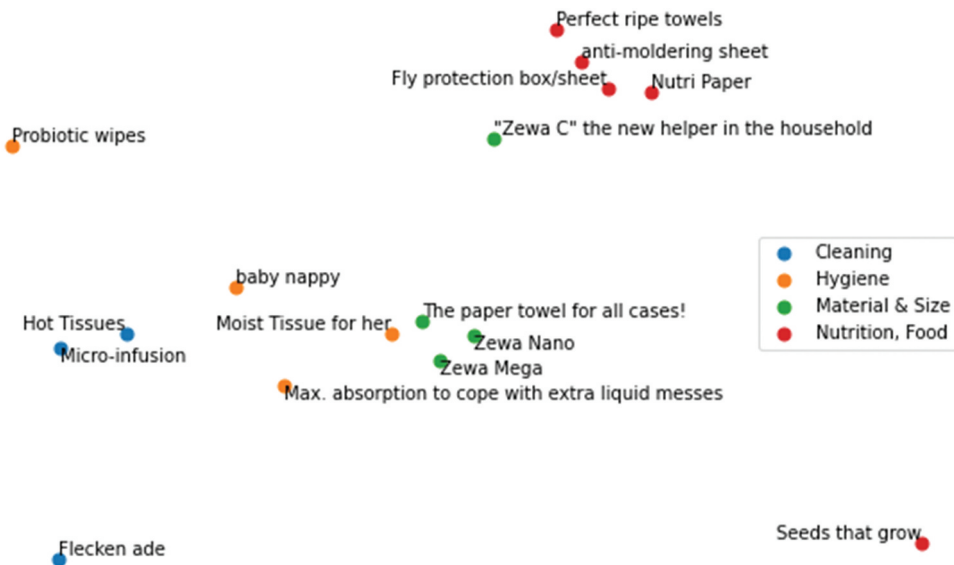


Figure A3. Cutout of crowdsourced idea space generated with SBERT embeddings.

Table A1. Descriptive statistics of averaged human novelty ratings per idea.

	Mean	Std	Min	25%	50%	75%	Max
Expert (n = 58)	3.01	1.14	1	2	3	4	5
Researcher (n = 225)	3.50	1.27	1	2.67	3.33	4.33	7
Prolific (n = 203)	3.70	0.59	2.11	3.36	3.78	4.13	4.78

Table A2. Examples of ideas and human evaluations.

Idea description	Expert	Researcher	Prolific
We touch office desks, chairs, car door handles, door knobs, window knobs, car seats, steering wheels, bench, surfaces etc. every day. There are no small towels (containing eco friendly disinfecting solution) to disinfect all these items, attachments and surfaces/areas that we touch daily. Create small towels, pre-absorbed with non staining, eco friendly disinfectant. Customers can use these small towels to disinfect small objects, surfaces and areas mentioned in pain point. The packaging should be single and small so that many can be carried in pockets and purses on daily basis. These towels should be use and throw, single use only. The disinfectant should be able to kill at least 99.999% germs, bacteria, virus etc. It should be very effective in killing COVID-19 bacteria. Otherwise no one will buy it. an example: ordinary wet wipes are normally alcohol free. Also they may contain extra ingredients like aloe vera, vitamin e, perfume etc. Basically a skin care product. Here the idea is to have a germ killing, especially COVID-19 virus and bacteria, solution soaked paper towel. Even if it is to be soaked in alcohol. Idea is more on effectiveness of its actual use than on skin care or perfumes.	1	1.67	2.57
Not so much of a pain point for customers, but more of a pain point for brands: how to keep pushing the boundaries when it comes to sustainable sourcing of the box materials and the roll core materials. When a brand is unable to alter the composition of the paper towels significantly, because the brand wants to maintain a familiar softness/strength/texture in the paper towel itself, then it's useful to look at the non-critical "accompaniments", i.e., the box, or the core of the roll, or the wrapping or shrink-wrap around the product. Company has scope for using recycled paper, waste materials, grasses, waste plant fiber, mushrooms, seaweeds, etc, or various combinations of them, to produce the box, core, wrapping.	2	2	2.71
Sometimes we need paper towels with a higher absorption power. Sometimes we need to cover a large area with paper towels for different activities. My proposal: 4 in 1 paper towels, with higher absorption power in 4 layers. These could be used like a regular paper towel or could be unfolded to cover large surfaces (nail paper surface, tablecloth, etc.). The edges will be stuck very subtly so that the layers will not come off until the client intends to do so.	2.33	3.67	3.9
There are dangerous pathogens lurking everywhere in your homes, you may not be able to see it but it can cause serious infections and make you terribly sick or worst case death. Sometimes you might have even wiped it out with your bare hands using a kitchen towel, so how do you know if you have touched or wiped something dangerous that makes you more cautious and reminds you to seriously wash your hands rather just an eye wash! A pathogen detecting towel, disposable or multi use it changes color after coming in contact with pathogens indicating danger.	4	6	4.64
Inspired by the idea that damp diapers can send out a signal tone, there is also a growing target group for cloths, toilet paper or all cleaning and cleaning utensils: general, dirt sensors can detect dirt, viruses, mold, spores. . . By RFID chip a signal is sent to the nearby receiver device. It would be possible to solve this via an app, because smartphones have become an extended body part of us anyway. :-). Maybe the individual hygiene defects could even be personalized, e.g., mold has clay 1, dust, sound 2, urine clay 3 etc. usf. This could also be extended with odor sensors, e.g., urine stains of animals or people can be cleaned for example with natural home remedies such as cloth, vinegar or lemon juice. There it would be interesting for people whose sense of smell is not (more) good, if sensors point out, here it still smells like "pissse", nicotine, fat. . . (I pre-arranged airing I might presume) Also thoughtable would be in a further development, that the signal tones will vary and is displayed or to be heard about what kind of hygiene deficiency it is, e.g., bacteria, fat mould is not used in combination.	5	6.33	4.4

Table A3. Pairwise correlations of algorithm-generated novelty scores.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Doc2vec	1st NN company (1)	1												
	1st NN scraped (2)	0.51	1											
	1st NN unique (3)	-0.03	-0.01	1										
	3rd NN company (4)	0.84	0.55	-0.04	1									
	3rd NN scraped (5)	0.56	0.83	-0.03	0.65	1								
	3rd NN unique (6)	0.03	-0.02	-0.04	0.08	0.09	1							
	LOF company (7)	0.87	0.49	-0.05	0.93	0.59	0.13	1						
	LOF scraped (8)	0.12	0.24	-0.05	0.14	0.23	-0.05	0.08	1					
	LOF unique (9)	0.08	0.06	-0.05	0.1	0.17	0.86	0.1	-0.02	1				
	1st NN company (10)	0.19	0.02	-0.03	0.24	0.01	0.06	0.24	0.01	0.04	1			
SBERT	1st NN scraped (11)	0.19	0.13	-0.07	0.21	0.07	-0.13	0.2	0.18	-0.08	0.64	1		
	1st NN unique (12)	-0.08	0	-0.11	-0.06	-0.02	0	-0.05	0.14	-0.02	0.02	0.05	1	
	3rd NN company (13)	0.24	0.04	-0.04	0.3	0.01	-0.01	0.3	0.04	0	0.85	0.66	0.03	1
	3rd NN scraped (14)	0.22	0.14	-0.05	0.25	0.07	-0.12	0.24	0.19	-0.08	0.7	0.96	0.06	0.72
	3rd NN unique (15)	0.2	0.05	-0.05	0.25	0.03	0.05	0.24	0.11	0.07	0.78	0.75	0.01	0.8
	LOF company (16)	0.23	0	-0.07	0.27	-0.05	0.05	0.26	0.06	0.05	0.76	0.62	0.04	0.89
	LOF scraped (17)	0.08	0.05	0.16	0.03	0.03	-0.08	0.08	0.17	-0.12	0.12	0.38	0.07	0.05
	LOF unique (148)	0.16	0.01	-0.09	0.18	-0.01	0.06	0.18	0.1	0.08	0.69	0.66	-0.03	0.68
	1st NN company (19)	0.27	-0.01	-0.04	0.31	0.02	0.06	0.31	-0.05	0.05	0.69	0.47	0.03	0.61
	1st NN scraped (20)	0.24	0.17	0.05	0.26	0.12	-0.1	0.25	0.16	-0.07	0.53	0.74	0.09	0.55
GPT-3 Ada Similarity	1st NN unique (21)	-0.06	0.06	0.05	-0.05	0.04	-0.1	-0.03	0.05	-0.01	0.07	0.13	0.05	0.07
	3rd NN company (22)	0.34	0.07	-0.05	0.4	0.12	0.02	0.4	0.06	0.05	0.65	0.48	0.05	0.7
	3rd NN scraped (23)	0.25	0.14	0.05	0.27	0.12	-0.1	0.27	0.14	-0.06	0.57	0.75	0.1	0.59
	3rd NN unique (24)	0.21	-0.01	0.01	0.25	0	0.17	0.25	0.11	0.18	0.61	0.54	0.03	0.62
	LOF company (25)	0.34	0.05	-0.02	0.38	0.07	0	0.38	0.06	0.02	0.65	0.53	0.07	0.7
	LOF scraped (26)	0.22	0.17	0.09	0.21	0.14	-0.11	0.22	0.14	-0.09	0.42	0.59	0.13	0.42
	LOF unique (27)	0.15	-0.04	-0.03	0.18	0	0.25	0.18	0.11	0.23	0.54	0.46	0	0.51
														0.5

(Continued)

left describes ideas addressing covers and stain protection, while the one on the orange at the top right towel holders are covered.

Human novelty evaluation

The following pre-existing solutions were presented by the host company in the contest briefing and had been exposed to the human novelty raters. They were also used as reference sets for the algorithm-generated scores.

- Create standard rolls and extra-long rolls for those who need larger sheets of paper for their application.
- Make household towels which come in nice and engaging designs.
- Make paper towel rolls with smaller half sheets to save paper.
- Create a nicely designed softbox dispenser to hold paper towels. They are hygienic and easy to dispense sheet by sheet.
- Make the paper kitchen towels absorbent when wet and especially strong so that you can easily wipe clean and scrub with them.
- Reduce the waste and CO2 emissions by increasing recycled content in all packaging.
- Reduce the waste and CO2 emissions by replacing fibres with fast renewable resources like straw.
- Reduce the waste and CO2 emissions by reducing the amount of fibres with thinner sheets.
- Reduce the waste and CO2 emissions by using recycled fibres instead of new wood fibres.
- A paper towel roll without a core to save cardboard.

Novelty detection with thresholds

In Figure 3, all the crowdsourced ideas are represented in a scatterplot based on their LOF scores relative to the pre-defined existing solutions (x-axis) and LOF uniqueness scores relative to the

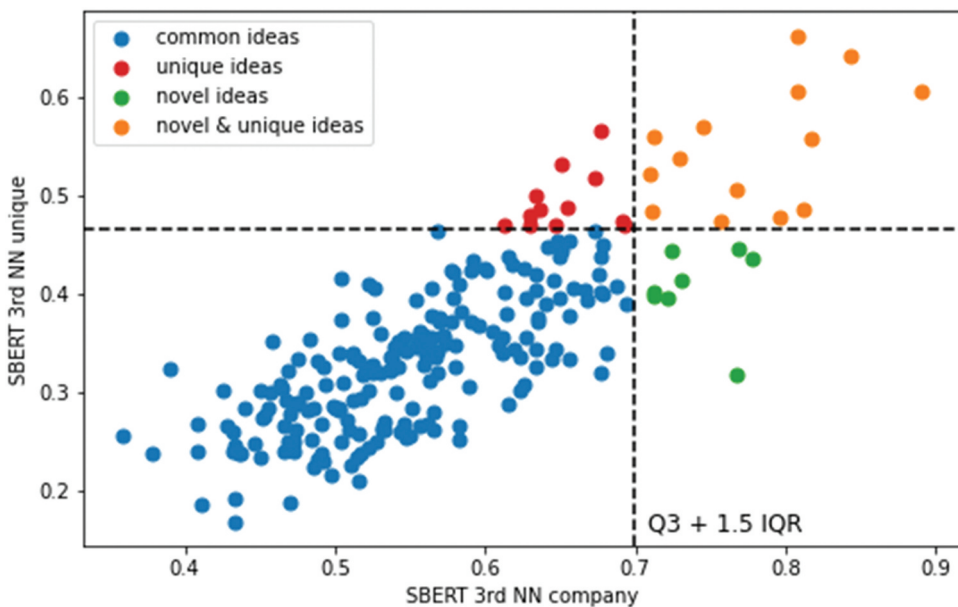


Figure A4. Combined mapping of novelty and uniqueness scores.

other crowdsourced ideas (y-axis). The scatterplot illustration allows to combine both, the novelty and uniqueness scores obtained through the LOF measure, and gain additional insights compared to analyses that only consider one measure. The threshold value was selected due to its objectivity and widespread application to detect deviant observations in statistical analyses. It calculates the sum of the third quartile and the by 1.5 multiplied InterQuartile Range ($Q3 + 1.5 \cdot IQR$). For example, novel and popular ideas may herald a new customer need or potential market that requires more rapid implementation of an idea. Conversely, novel and unique ideas may indicate interesting solutions, including some question marks about the actual potential, which require further elaboration.